COMPACT PHOTO MANUAL

ESP32-C3 LED Matrix

Assembly & setup

This guide covers assembly and first-time setup. Gather the printed parts, LED matrix, ESP32-C3, three jumper wires, USB data cable and project software.

RED = power
WHITE = ground
GREEN = data
The controller jumper shown uses BLACK for ground; it connects to the WHITE matrix ground.

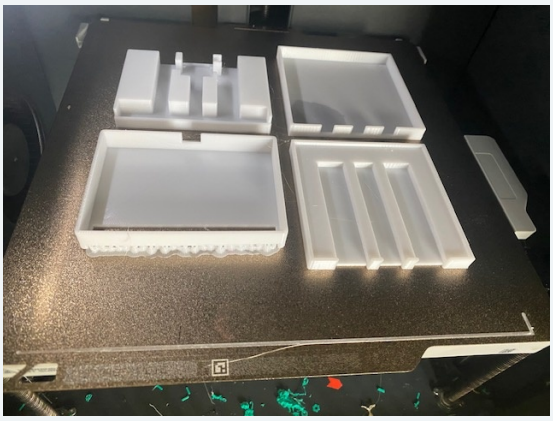
Work with power disconnected. Wear eye protection when cutting, ventilate solder fumes and keep the hot iron away from printed plastic. Let joints cool before fitting the board.

Before power-on: verify supply voltage/current, polarity, common ground, data input and the program GPIO. Check solder bridges and insulate each unused wire end separately.

Power safely. Use the correct supply for the project. Do not use GPIO as a power connection. Avoid connecting USB and a separate supply together unless the board supports it.

Steps 01-02

01. Check the printed parts



Lay out the printed parts shown: matrix holder, front screen/diffuser, base and cover pieces. Check for warped edges, loose strands and blocked openings.

PHOTO: all printed

02. Remove the support



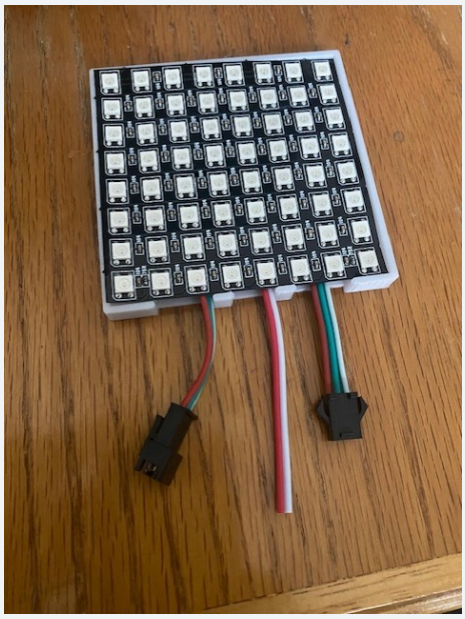
Remove the sacrificial support along the long opening shown. Work in small sections while holding the finished part securely.

Trim remaining strands and lightly smooth rough mating edges. Keep tools pointed away from your hands. Preserve the opening and retaining edges.

PHOTO: remove support

Steps 03-04

03. Fit the matrix into its holder

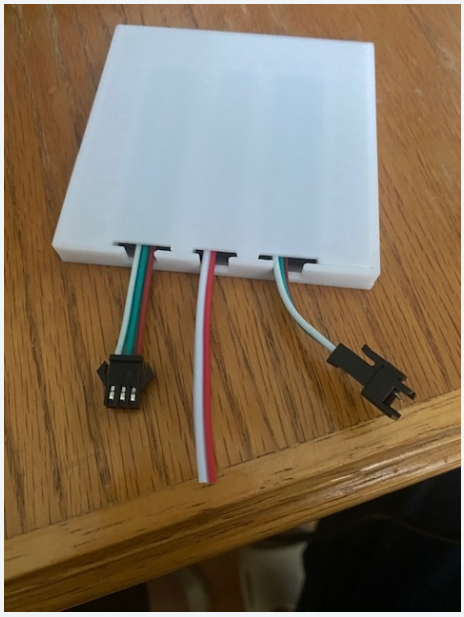


Orient the LEDs toward the open front of the holder. Route the three cable groups through the bottom slots as shown.

Press gently around the board edges until it sits evenly. Avoid pressing on individual LEDs or bending the board. If tight, remove it and correct the fit.

PHOTO: squeeze fit to holder

04. Add the front screen



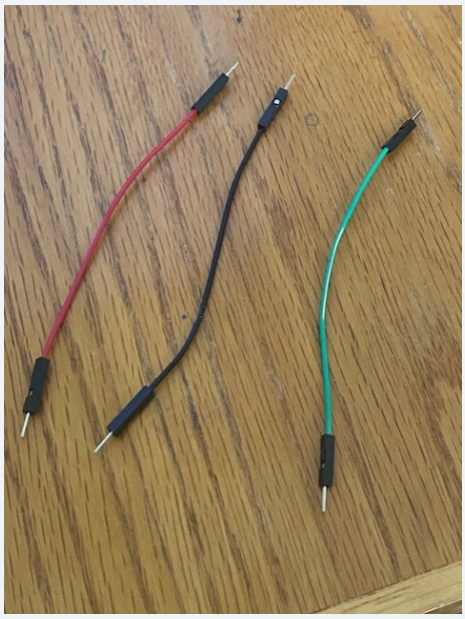
Use a WHITE front screen/diffuser so it does not tint or distort the intended LED colors. Place it over the LED face and align it with the holder.

Keep each cable group in its own exit slot. Seat the cover evenly and check that no wire is pinched.

PHOTO: put on screen

Steps 05-06

05. Cut and prepare the jumper wires

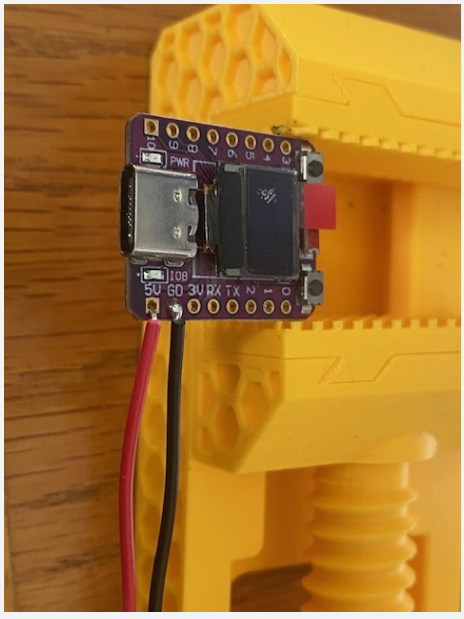


Measure the route from the controller to the matrix connector. Cut the RED power, ground and GREEN data jumpers to length, leaving a little slack and retaining a male pin at the connector end.

Strip a short length of insulation from each cut end, twist the strands and tin them ready for soldering. The photo uses BLACK for the controller ground jumper; it connects to WHITE matrix ground.

PHOTO: prepare 3 wires with pin

06. Solder power and ground



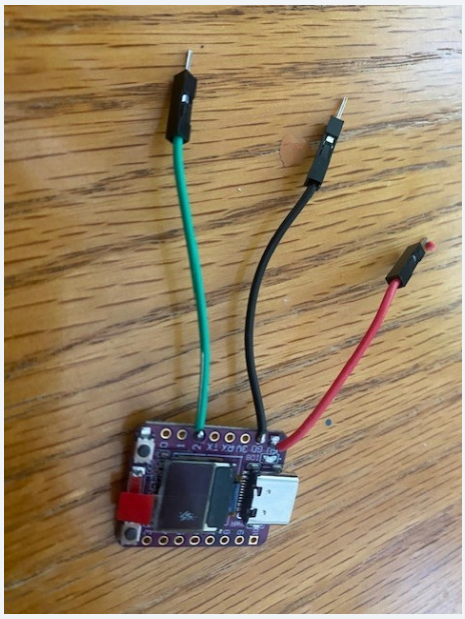
With power disconnected, solder RED to the **5V** pad and BLACK to ground as shown. Keep the wires clear of neighboring pads.

Tin each wire and pad, heat briefly, then hold still while cooling. The green data wire is added next.

PHOTO: solder power ground and data

Steps 07-08

07. Check the completed wiring

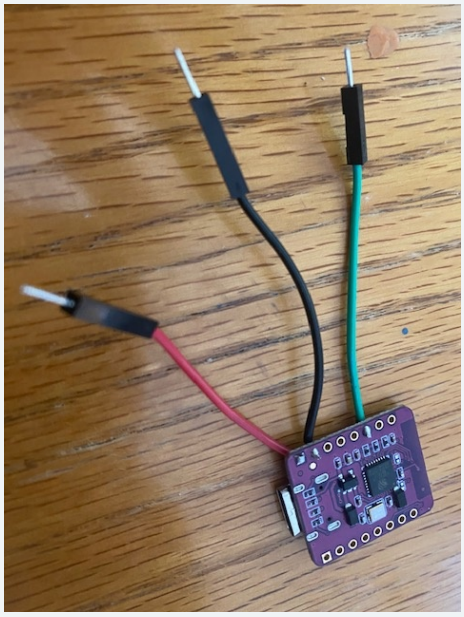


RED is at 5V; BLACK is at the adjacent ground pad. Inspect the joints for bridges to neighboring pads.

GREEN appears at pad 2, immediately after TX. The project program uses LED data on **GPIO 2**. Verify the board label, inspect for solder bridges and gently tug each lead.

PHOTO: solder complete

08. Inspect both board faces



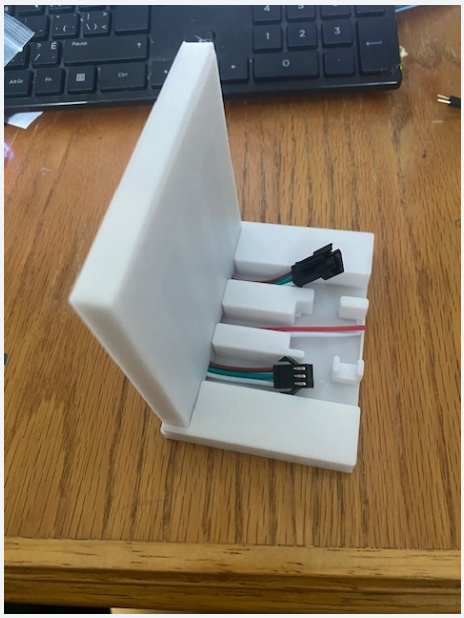
Turn the board over and check the reverse face of the three connections. Ensure solder wets the pad and conductor without touching neighboring pads.

The caption says "do both sides"; this is the same board from the rear. Touch up only a joint that needs it. Do not create extra connections or overheat the board.

PHOTO: do both sides

Steps 09-10

09. Seat the panel in the base

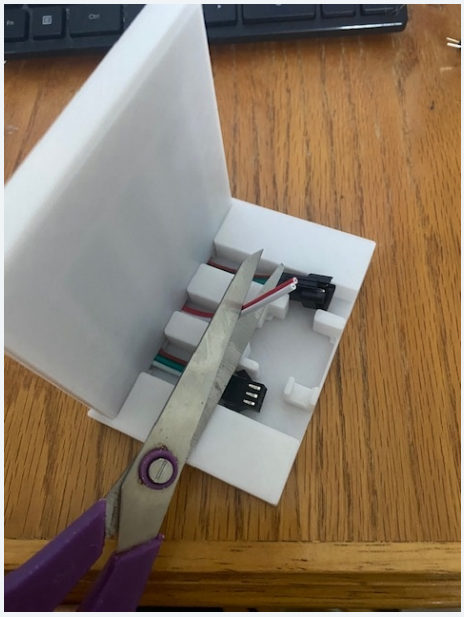


Fit the completed panel into the upright slot in the base. Guide its cables through the spaces between the printed ribs.

Leave enough slack to reach the controller location and keep the USB port accessible. Do not trap the cables underneath the panel.

PHOTO: put into base

10. Remove the unused wire pair



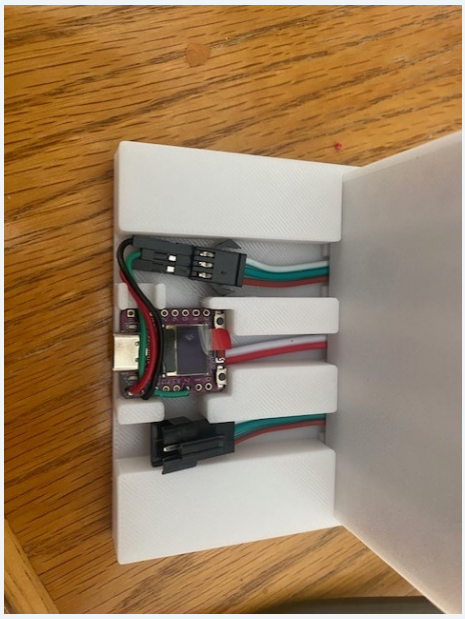
The separate central RED/WHITE pair is not needed for this build. Cut it off as shown, with all power disconnected.

Cut one conductor at a time. Insulate each remaining end separately so bare power and ground cannot touch.

PHOTO: cut unneeded wires

Steps 11-12

11. Connect the matrix input



Connect **RED power, WHITE ground and GREEN data**. The BLACK controller jumper connects to the WHITE harness ground.

Use the matrix DATA INPUT connector. Match each wire to the correct contact, insert the pins fully and secure the connections. Keep the unused output connector clear.

PHOTO: stick pins in holes red is power white is ground and green is data

12. Fit the base cover



Place the controller in its recess with the USB port at the opening. Fold wire slack gently into the channels; keep connectors away from the cover edges.

Lower the base cover evenly. If it will not seat, lift it and reroute the wires. Complete the polarity and short-circuit inspection before closing.

PHOTO: put on cover base

Steps 13-14

13. Check the finished assembly

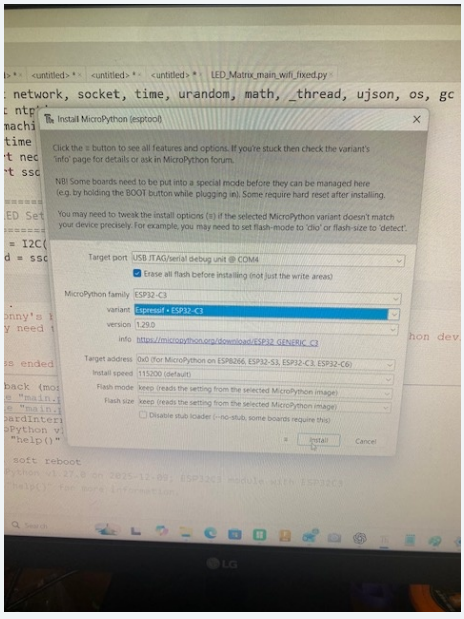


Check that the upright panel is stable, the front screen is seated, and the base cover closes without pressure on wiring.

Confirm that the USB opening remains accessible. Keep the unit unpowered until the voltage, ground, data input and supply ratings have been checked.

PHOTO: fully assembled

14. Install MicroPython via Thonny

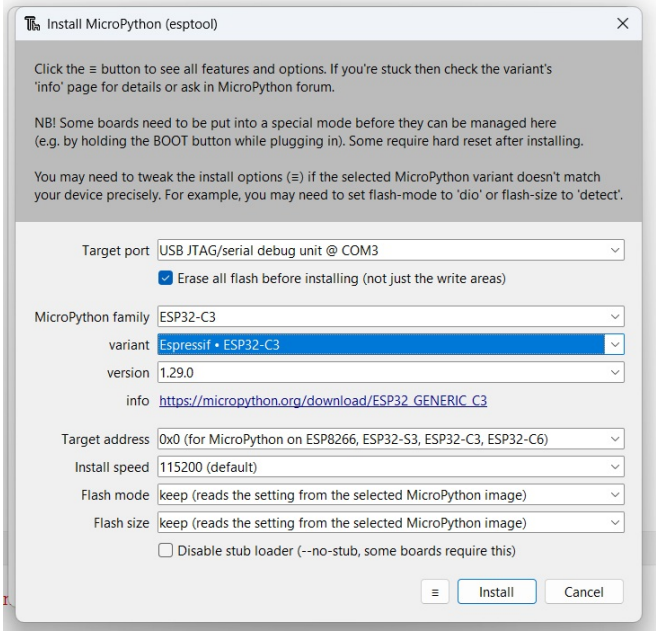


Download and install Thonny from **https://thonny.org/**. Connect the board by USB data cable. Select MicroPython (ESP32), then Install or update MicroPython.

Choose the device port and compatible ESP32-C3 firmware. The photo shows v1.29.0 and COM4; the device port may differ. Back up files before erasing flash. Follow the software pages next.

PHOTO: install thonny to esp32c3

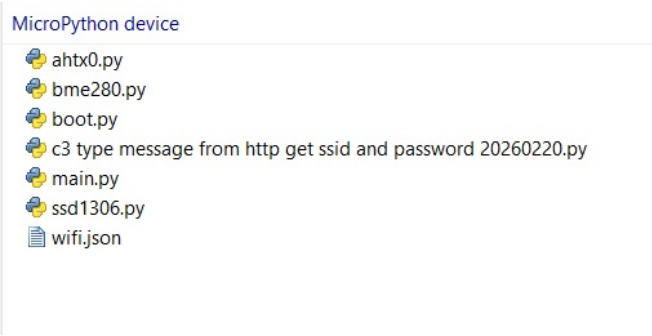
Install firmware & files



1. Install Thonny, then firmware. Download Thonny for the computer from **https://thonny.org/**.

Select MicroPython (ESP32), then Install or update MicroPython. Choose the device port and ESP32-C3 firmware. The screenshot shows v1.29.0 and COM3; the device port may differ.

Back up files before erasing flash. Follow the board boot procedure if needed.



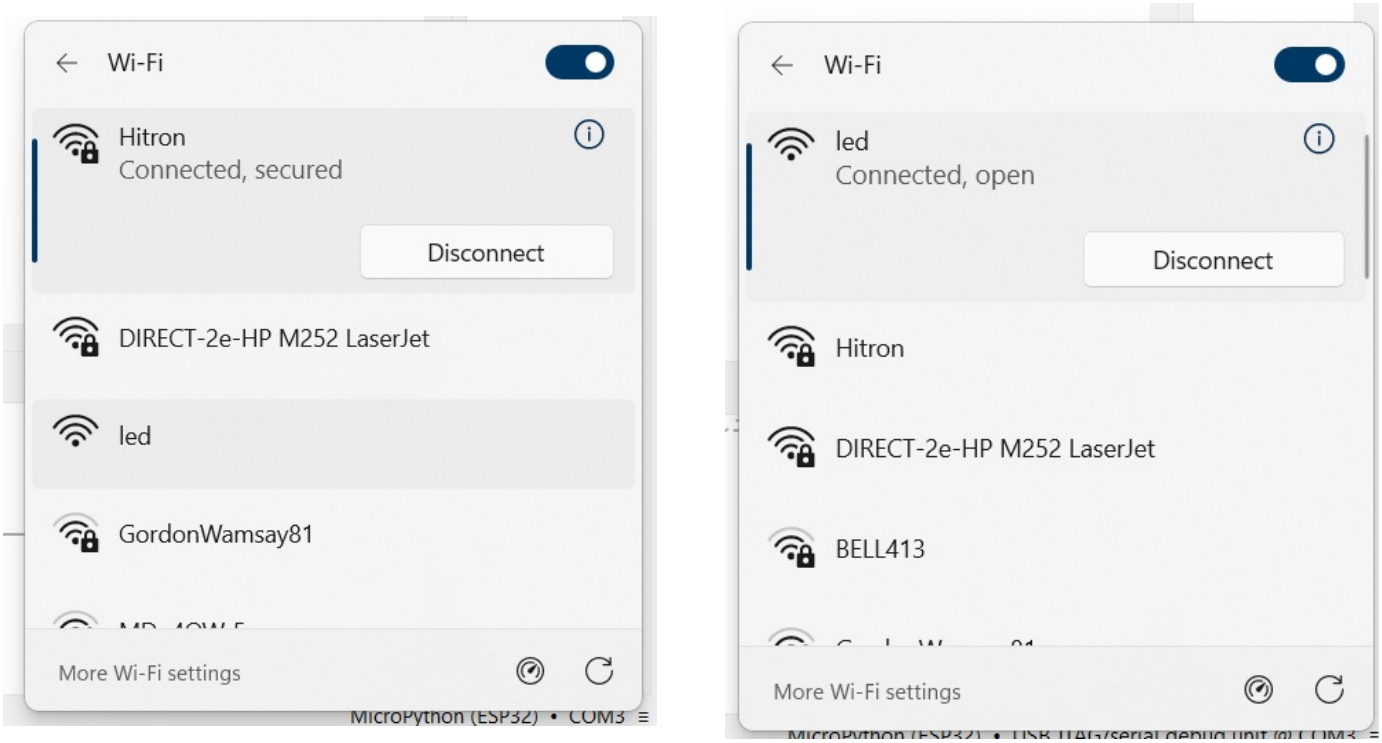
2. Copy the four ZIP files. Extract main.py, ssd1306.py, ahtx0.py and bme280.py. In Thonny open each file, use Save as, and select the MicroPython device root.

The screenshot contains extra files from an existing board. boot.py and the long-named script are not in the ZIP and do not need to be added. wifi.json is created after setup.

3. Restart. Select the correct interpreter and port. The program uses GPIO 2 for LED data and GPIO 6/5 for OLED clock/data.

Thonny firmware help | ESP32-C3 firmware requirements

Join the led network



LEFT: led is available. **RIGHT:** led is connected and open.

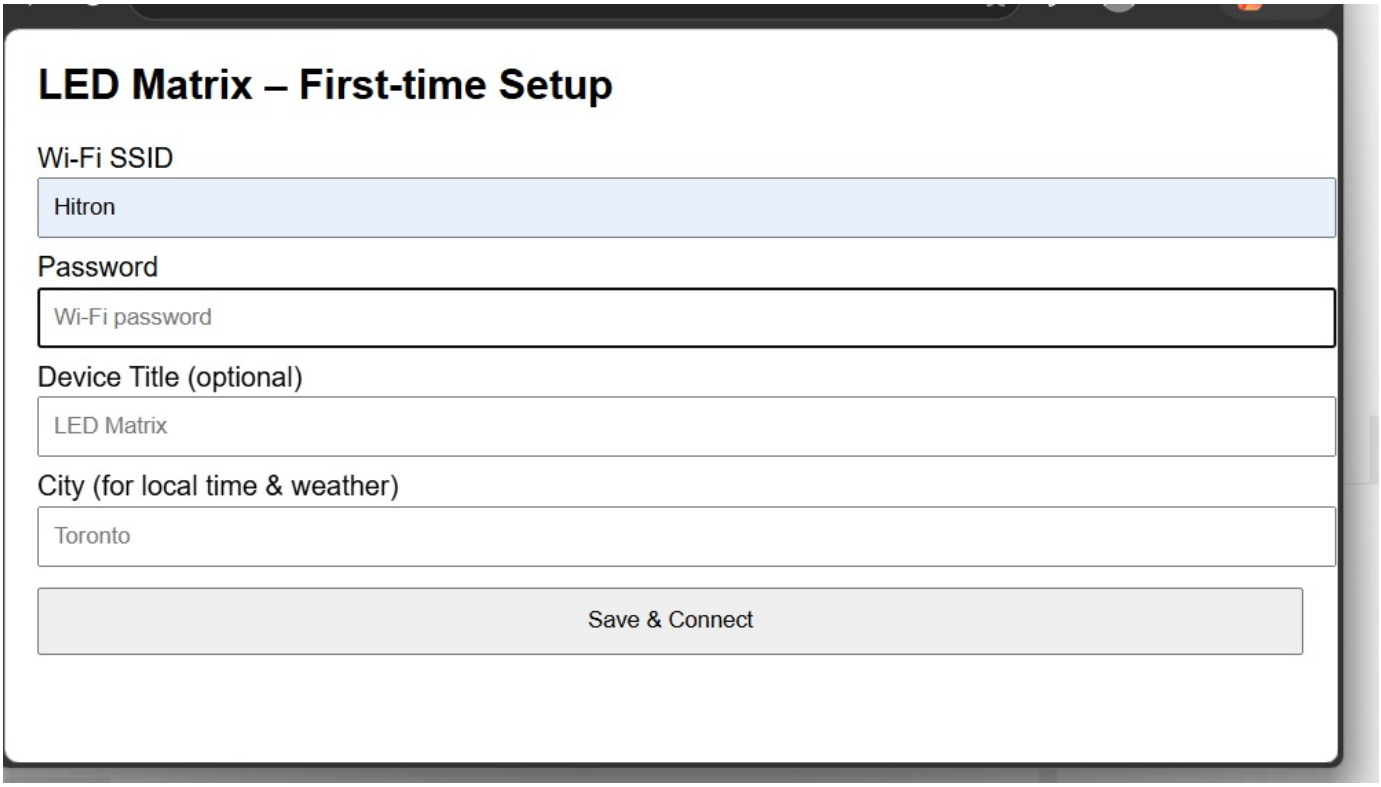
1. Open Wi-Fi settings on the phone or computer and select **led**. It has no password. Stay connected if warned there is no internet.

2. Confirm the connection. The network should show connected. Hitron is the example home network in these screenshots; use the home network later.

3. Open a browser and type **http://192.168.4.1** in the address bar. Use HTTP. The next page shows the setup form.

If led is absent, check that main.py and ssd1306.py are on the board and restart. A board already connected using saved Wi-Fi settings does not open this setup network.

Enter Wi-Fi, city & name



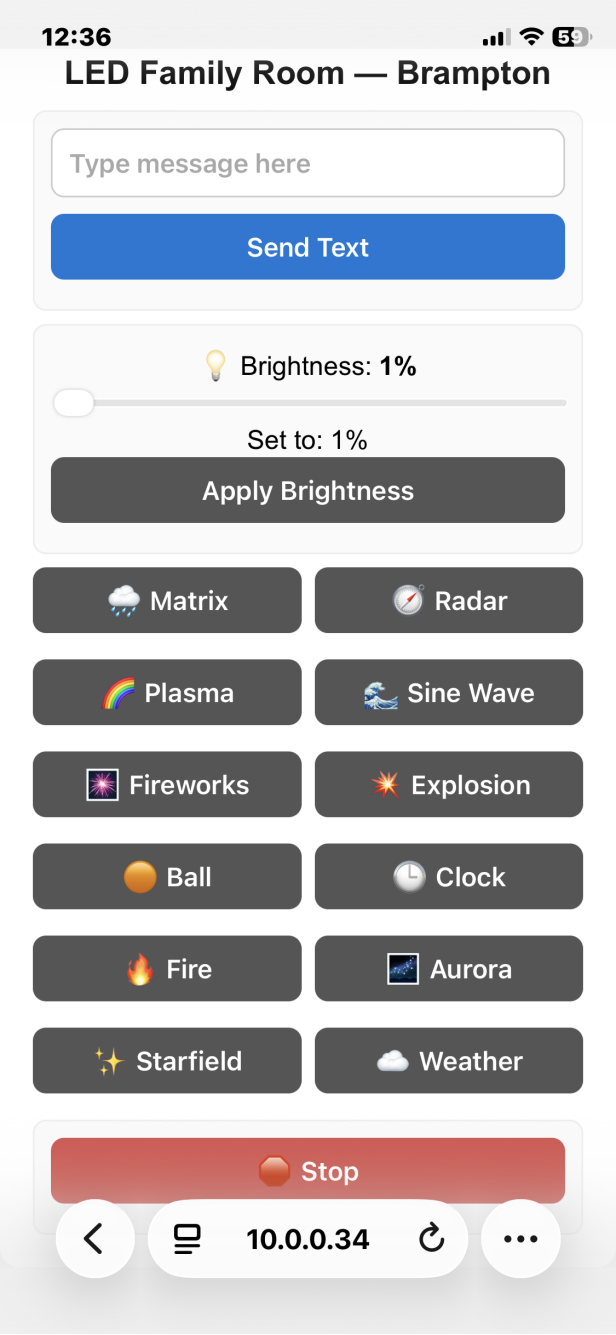
Wi-Fi SSID: the home 2.4 GHz network name.
Password: that network's Wi-Fi password.
Device Title (optional): a device name, such as LED Family Room; blank uses LED Matrix.
City (for local time & weather): enter the local city, such as Brampton.

The grey LED Matrix and Toronto text in the screenshot is placeholder text. Type the intended values. Hitron is only an example SSID.

Select **Save & Connect**. SSID and password must both be filled in. The page reports "Saved. Trying to connect..." The program creates wifi.json, closes led and tries the home Wi-Fi. No manual restart is needed.

Reconnect the phone or computer to home Wi-Fi. Open the new IP shown on the OLED, scrolled on the matrix or printed in Thonny as "Web UI: http://.../".

Use the control page



Open the device address.
The example shows 10.0.0.34. Use the IP shown by the unit, not 192.168.4.1 after setup.

Send Text: type a message and tap the button.

Apply Brightness: move the slider, then apply. Start low.

Modes: choose an animation, Clock or Weather.

Stop: ends the active display.

The title shows the saved device name and city. The IP can change after reconnecting.

Later power-ups reconnect using wifi.json. Internet access is needed for city lookup and online time/weather services.



If setup does not work

Setup page will not open: confirm you are still connected to led, then type http://192.168.4.1 directly. If needed, temporarily disable mobile data or a VPN that routes local traffic elsewhere.

Save returns to the form: this program requires a nonempty SSID and password. It does not accept an open home network with a blank password.

Correct credentials still fail: this version decodes spaces only in form values. Special characters such as &, +, % or # can be saved incorrectly after browser encoding. A program update may be needed to handle these characters correctly.

Wrong city or time: the program uses the first city-search result. Enter a recognizable city and check the Thonny "Geocoded" message. If lookup fails, its built-in fallback is Toronto. For a different city after initial setup, back up wifi.json in Thonny, delete it from the device and restart; repeat setup. The backup contains the saved Wi-Fi password.

Dark matrix or repeated resets: disconnect power and check supply capacity, polarity, common ground, matrix data input and GPIO 2. Disconnect immediately if wiring or components become unusually hot.

Hardware: this guide uses an ESP32-C3 OLED board and an 8 x 8 RGB LED matrix with the project software.